

THE OPERATIONAL BRAIN FOR FOUNDER-LED SMES

DECISIONOS

PROJECT DOCUMENT

Speak the decision. We run the company. — an AI operating system that turns spoken directives into structured tasks, workflows and a shared company brain.

Complete product walkthrough · architecture · features · API · screens (desktop & mobile)

DECISIONOS — COMPLETE PROJECT DOCUMENT

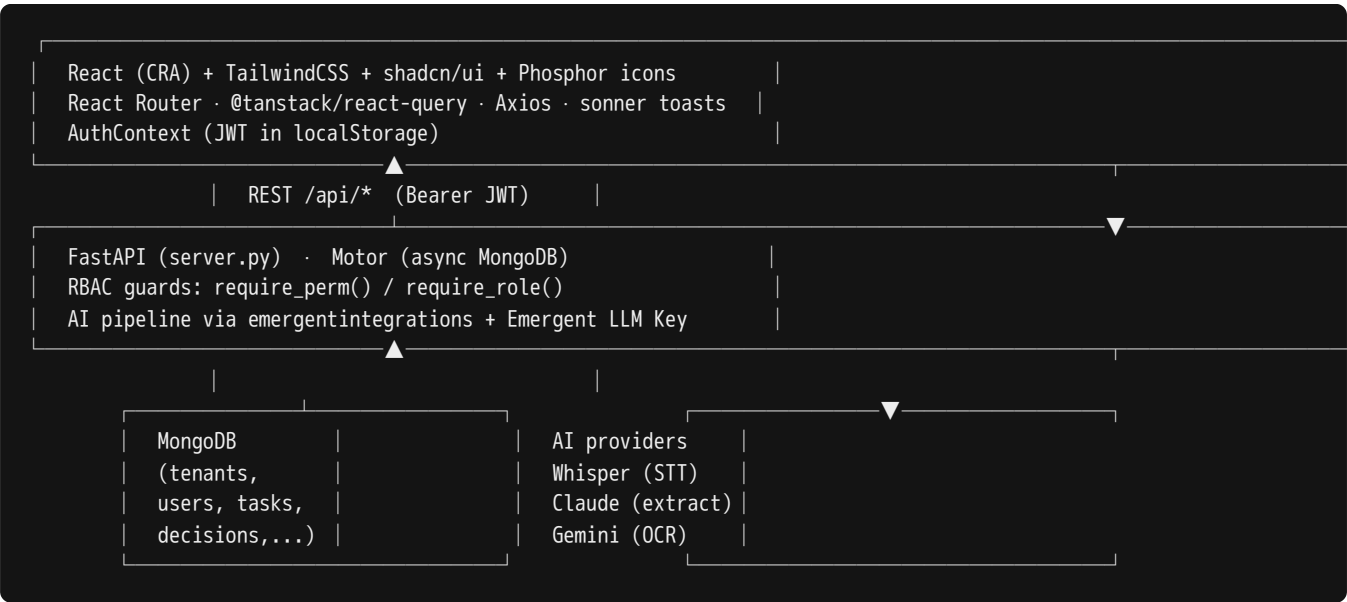
Tagline: *Speak the decision. We run the company.* **What it is:** An AI operating system ("operational brain") for founder-led SMEs. Founders capture decisions by voice or text; AI turns them into structured, assigned, trackable tasks and workflows. Everyone works from one shared source of truth.

This document explains the end-to-end product: architecture, the complete user flow, every screen/feature, the data model, the API, authentication, AI/integrations, and what's live vs. pending. Desktop and mobile screens are shown in the chat alongside this document.

1. Product Overview

Who it's for	Founders / owners of small & mid-size businesses (the demo is <i>Sharma Textiles Pvt Ltd</i> , a textile manufacturer) and their teams (Sales, Production, Finance, and custom roles).
Core promise	A founder speaks or types an instruction → AI extracts the decision, creates tasks, assigns owners, sets deadlines, and tracks execution — surfacing "fires" the founder must handle.
Model	Multi-tenant SaaS. Each company = one tenant . Users belong to a tenant with a role and granular permissions (module-level RBAC).
Design language	Neo-brutalist: sharp black borders, hard shadows, red (brand-red) + yellow + blue accents, monospace labels, high-contrast. Fully responsive (desktop sidebar + mobile bottom-nav & drawer).

2. Tech Stack & Architecture



- **Frontend** talks to backend only via `REACT_APP_BACKEND_URL` and every route is prefixed `/api`.
- **Backend** uses `MONGO_URL` / `DB_NAME` from env. All datetimes are timezone-aware UTC.
- **Auth:** JWT Bearer tokens; three sign-in methods (see §6).

- **AI:** OpenAI Whisper (speech → text), Claude Sonnet (decision/task extraction), Gemini Vision (document/PDF OCR) — all through the Emergent LLM Key.

Frontend structure

```
frontend/src/
├── context/AuthContext.js      # login / register / loginWithOtp / refreshTenant
├── lib/perms.js                # PERMISSIONS, role defaults, hasPerm(), userPerms()
├── components/Layout.js        # sidebar (desktop) + drawer & bottom-nav (mobile)
├── pages/
│   ├── Login.js                # sign in / 7-step onboarding / invite deep-link
│   ├── Inbox.js                # "Decision Desk"
│   ├── CEOBrief.js             # CEO Brief / My Brief
│   ├── MyWork.js + Tasks.js    # My Work board + AI Priority
│   ├── People.js (Team.js + Contacts.js) # Employees / Customers / Vendors
│   ├── Brain.js (+AskAI.js)    # Company Brain: Ask + Search
│   ├── Ingest.js               # "Capture" data ingestion
│   ├── Workflows.js            # Order Fulfilment / Procurement kanban
│   ├── Meetings.js             # Meeting Notes
│   ├── OperatingScore.js       # Owner-only company health
│   └── WorkCoach.js, Journal.js, Notifications.js, ContactProfile.js
```

3. Roles & RBAC (module-level access)

Every screen and API is guarded. Owners see everything; employees see only the modules their permissions allow. Blocked routes render a dedicated **Access Denied** view.

Permission keys (`lib/perms.js` ↔ backend `require_perm`): `inbox` (Decision Desk), `people`, `brain`, `ask`, `data_input` (Capture), `workflows`, `team_manage`.

Always available to everyone: CEO Brief (role-aware), My Work, Meeting Notes. **Owner-only:** Operating Score, voice/text capture on Decision Desk, CEO Journal.

Backend enforcement examples: `contacts→people`, `workflows→workflows`, `brain/ask→brain/ask`, `ingest→data_input`, `operating-score→owner`, `/users→team_manage`.

4. The Complete User Flow

4.1 Onboarding (new workspace) — `Login.js`, 7 steps

1. **About company** — name, your name, work email, password, mobile, industry, GST, branches.
2. **Business scale** — team size, currency, monthly sales/purchases, # customers/suppliers, region.
3. **Current software** — Excel / Tally / Zoho / Others.
4. **AI-suggested team & offerings** — AI proposes roles + products for your industry (`POST /api/onboarding/suggest`); edit freely, then **Create workspace** (`POST /api/auth/register`).
5. **Connect data** — upload Excel/CSV (customers, invoices, payments) via the ingestion pipeline; Tally/Zoho shown "coming soon".
6. **Invite team** — add employee mobile numbers.
7. **AI learns your business** — animated setup, then enter DecisionOS.

4.2 Daily loop (owner)

```
Speak/type a directive (Decision Desk)
| Whisper → Claude extraction
▼
Decision + Tasks created (assignee, role, due date)
|
▼
Approve / Reject on Decision Desk → Tasks land in team members' "My Work"
|
▼
CEO Brief shows counts + "fires to put out" ← drill-downs into source screens
|
▼
Send Daily Digest / review Operating Score
```

4.3 Employee loop

Sign in (password or mobile OTP) → **My Work** (AI Execution Guide, complete tasks, upload photo/voice proof, escalate/handoff) → **My Brief** (personal CEO Brief + AI Coach) → **Meeting Notes**.

5. Screen-by-Screen Feature Guide

Screens below correspond to the desktop and mobile screenshots shared in chat.

5.1 Login / Sign-in (*desktop + mobile shown*)

- **Email & Password** tab and **Mobile OTP** tab.
- One-tap **Sharma demo** buttons (Owner / Sales / Production / Finance).
- Handles **invite deep-links** (`/login?invite=<token>`) — see §6.3.

5.2 Decision Desk (Inbox) (*desktop + mobile shown*) — perm `inbox`

- **Tap to speak a decision** (Auto / EN / தமிழ் / Tanglish language modes) + **Type a directive** box → **Structure it**.
- **Decision Approvals**: AI-structured directives awaiting Approve/Reject, each showing the auto-assigned team/role; "Add team member / task" inline.
- **Tasks & Activity** feed with filters (Customer, Supplier, Invoice, Payment, Complaint, Task, Approval, Reminder) and uploaded-document cards.
- Employees see a note that capture is owner-only but still get the company inbox feed.

5.3 CEO Brief / My Brief (*desktop + mobile shown*) — always available (role-aware)

- Period toggle: **Morning / Evening / Weekly / Monthly**.
- Clickable counter cards: **delayed tasks, completed, waiting for approval, employees absent, customer complaints, payments overdue** — each opens a drill-down modal with in-place actions (approve/reject, resolve complaint, review purchase) and deep-links into the source screen.
- **"Fires to put out today"** red banner for owners (escalation count).
- Owner: company-wide brief + **CEO Journal**. Employee: personal brief + **AI Coach**. Auto-refreshes every 30s.

5.4 My Work (*desktop shown*) — always available

- **My Work** vs **Board** (kanban) toggle; **AI Priority** toggle re-orders tasks by AI score (Impact/Revenue/Risk/Urgency).
- Per-task **AI Execution Guide** (generated step checklist with per-step "Ask AI" + "Update"), **Complete**, **Photo/Voice proof**, **Activity & Handoffs**, **Update / Escalate**.
- **Messages** rail: escalations addressed to you, owner alerts for overdue tasks.

5.5 People (*desktop + mobile shown*) — Employees perm `team_manage` ; Customers/Vendors perm `people`

- Tabs: **Employees / Customers / Vendors**.
- **Employees**: member list with role, access-count, **Access** editor (role + granular permissions + live "menus this member will see" preview), **Mark absent/present**, and **Invite** (generates a passwordless invite link — §6.3). **Add Member** dialog supports **Password** or **Mobile OTP (passwordless)** members.
- **Customers/Vendors**: CRM-lite contacts; each opens a **360° Contact Profile** with auto-calculated financials (invoices, payments, balance) and AI health re-scoring.

5.6 Company Brain (*desktop shown*) — perm `brain / ask`

- **Ask** tab: RAG chat grounded in your company data (`POST /api/ask`) with suggested prompts ("What purchases need my approval?", "Which tasks are overdue?", "Summarise open sales orders", "What did I decide about festive stock?").
- **Search** tab: deterministic keyword record browser (`GET /api/brain/search`) that traces any founder decision to the tasks/workflows it created.

5.7 Capture (Ingest) (*desktop shown*) — perm `data_input`

- **PDF / Photo** → OCR (Gemini Vision) extracts invoices/bills/receipts/POs.
- **CSV / Excel** → columns auto-detected → preview → commit (creates contacts/invoices/payments).
- **WhatsApp forwarding** — "coming soon" (same pipeline).
- **Recent uploads** list with per-file counts (Contacts / Invoices / Payments / Tasks) and FILED status.

5.8 Workflows (*desktop shown*) — perm `workflows`

- Two flagship pipelines as kanban boards:
- **Order Fulfilment**: Order Received → Confirmed → In Production → Ready → Dispatched → Delivered.
- **Procurement**: Purchase → Payment.
- Cards carry customer, amount, and an **Advance** action to move stages.

5.9 Meeting Notes — always available

- Record/transcribe a meeting → AI writes minutes: **Key Points**, **Decisions**, **Action Items** (assignable to members).

5.10 Operating Score (*desktop shown*) — owner-only

- **Company Health /100** with pillar bars: **Execution**, **Finance**, **Sales**, **Responsiveness**.
- Headline metrics: Tasks Done, Open Tasks, Overdue, Open Complaints.
- **Team Execution** leaderboard (done/open/overdue per member) with per-member **Coach** button (opens their Work Coach).

5.11 Notifications — always available

- Reminders & escalations feed; mark-read / read-all; follow-up engine.

6. Authentication

6.1 Email + Password

`POST /api/auth/login` → JWT. `POST /api/auth/register` creates the tenant + owner.

6.2 Mobile OTP login (*DEV mode*)

`POST /api/auth/otp/request` finds the user by last-10-digits of phone, generates a 6-digit code (hashed, 5-min TTL, 5-attempt max, 30s resend cooldown) and returns it as `dev_otp` (real SMS pending — see §8). `POST /api/auth/otp/verify {phone, code}` → JWT. Wiring for Twilio already exists; flipping the provider on requires only credentials.

6.3 Passwordless members + auto invite links (*new*)

- Owners can add an employee with **no password** (People → Add Member → **Mobile OTP** toggle). Such users have `passwordless: true`; email/password login is rejected — they sign in only via OTP.
- Adding a member (with a phone) returns an `invite_token`; the owner gets a **one-tap invite link** (`<origin>/login?invite=<token>`) with **Copy** and **WhatsApp share**, plus a per-row **Invite** button that regenerates a link (`POST /api/users/{id}/invite`).
- Opening the link auto-selects the OTP tab, shows a welcome banner (name / company / masked phone), texts the code (dev: auto-fills it), and one **Verify & sign in** lands the member in the app. Links expire in 7 days.
- Endpoints: `GET /api/auth/invite/{token}` (welcome info), `POST /api/auth/invite/{token}/start` (send OTP). `invite_token` is never leaked in `GET /api/users`.

7. Data Model (MongoDB, key collections)

COLLECTION	KEY FIELDS
tenants	name, industry, company_size, region, currency, gst, branches, roles[], products[], business_scale, invited_employees[]
users	id, tenant_id, name, email, phone, role, permissions[], password_hash, passwordless, invite_token / invite_expires_at
decisions	id, tenant_id, text, dtype, owner_id, status
tasks	id, tenant_id, title, description, status, assignee_id, assignee_role, due_date, steps[], execution_plan, updates[], parent_task_id, raised_by, priority_score
contacts	id, tenant_id, type (customer/dealer/vendor), name, company, phone, email, tax_id, tags[], status, assigned_id
invoices / payments	linked to contacts; power 360° profiles & financial pillars
workflows	pipeline, stage, customer, amount
meetings	transcript, key_points, decisions, action_items
otp_codes	phone, code_hash, expires_at, attempts
notifications, attendance, complaints, memory, ingestions,	supporting operational data

COLLECTION	KEY FIELDS
voice_notes	

8. AI & Third-Party Integrations — status

INTEGRATION	PURPOSE	STATUS
OpenAI Whisper	Speech → text (voice capture)	✅ Live (Emergent LLM Key)
Claude Sonnet	Decision & task extraction, Ask AI	✅ Live (Emergent LLM Key)
Gemini Vision	PDF/photo OCR in Capture	✅ Live (Emergent LLM Key)
SMS OTP delivery	Real invite/login codes	🕒 DEV mode — backend generates & verifies; SMS send mocked. Awaiting APM Technologies verify/send-SMS endpoint or Twilio keys.
WhatsApp Business Cloud	Inbound/outbound ingestion	🕒 Pending keys (webhook scaffolded)
Zoho Books	Customers/invoices/payments sync	🕒 Pending keys
Resend	Email daily digest	🕒 Pending keys
Tally	Read-only financials	🔮 Future (needs local agent)

APM Technologies note: the shared doc only exposes `/Registration` and `/ForgotPassword`, which generate APM's *own* OTP with no verify endpoint and no custom-message send — incompatible with our backend-verified OTP until APM provides a `/Verify` or a generic Send-SMS endpoint.

9. API Reference (grouped)

Auth & tenant `POST /api/auth/register` · `POST /api/auth/login` · `GET /api/auth/me` · `PATCH /api/tenant` · `POST /api/onboarding/suggest` `POST /api/auth/otp/request` · `POST /api/auth/otp/verify` `GET /api/auth/invite/{token}` · `POST /api/auth/invite/{token}/start`

Users & invites `GET/POST /api/users` · `PATCH /api/users/{id}` · `POST /api/users/{id}/invite` · `GET/POST /api/invites`

Capture / voice / meetings `POST /api/voice-notes` · `POST /api/voice-notes/text` · `POST /api/capture/clarify` · `GET /api/voice-notes[/]{id}]` `POST /api/meetings[/text]` · `GET /api/meetings[/]{id}]` `POST /api/ingest/document` · `POST /api/ingest/csv` · `POST /api/ingest/{id}/commit` · `GET /api/ingest[/]{id}]`

Decisions & tasks `GET /api/decisions[/]{id}[/timeline]` · `GET /api/journal` `POST /api/decisions/{id}/tasks|approve|reject` `GET/POST /api/tasks` · `PATCH /api/tasks/{id}` · `POST /api/tasks/{id}/execution-plan/generate` · `PATCH .../execution-plan` `POST /api/tasks/{id}/steps/ask|updates|respond|attachment` · `POST /api/tasks/prioritize`

Brain, brief, workflows, contacts, ops `POST /api/ask` · `GET /api/brain/search` `GET /api/brief` · `GET /api/brief/details` · `POST /api/brief/send-digest` · `GET /api/dashboard` `GET/POST /api/workflows` · `PATCH /api/workflows/{id}/advance` `GET/POST /api/contacts` · `PATCH/DELETE /api/contacts/{id}` · `GET /api/contacts/{id}/profile`

• POST .../rescore GET /api/invoices • GET /api/payments • GET /api/inbox • POST /api/inbox/{id}/status GET /api/operating-score • GET /api/work-coach • POST /api/work-coach/refresh GET /api/notifications • POST /api/notifications/{id}/read|read-all • POST /api/follow-up/run GET/POST /api/attendance • GET/POST /api/complaints • PATCH /api/complaints/{id}/resolve GET/POST /api/memory • GET /api/calendar • GET/POST /api/webhooks/whatsapp

10. Mobile Experience (*screens shown*)

- **Bottom tab bar:** Desk • Brief • Work • People • Brain.
 - **Hamburger drawer:** full menu + company details + **Send Daily Digest** (owner) + **Sign out** (clears the fixed bottom-nav overlap).
 - All screens are single-column responsive; cards, filters and drill-downs adapt to small viewports.
-

11. Demo Credentials

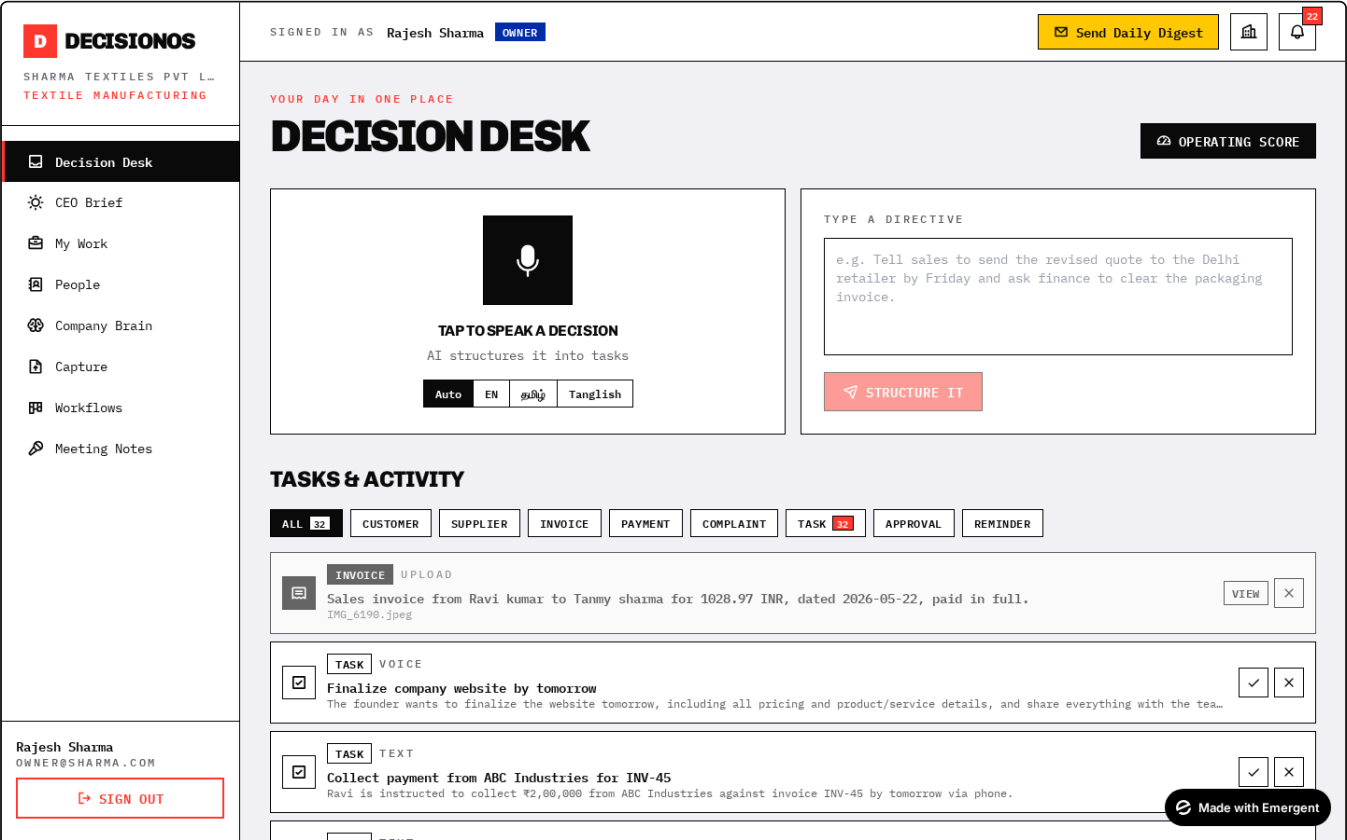
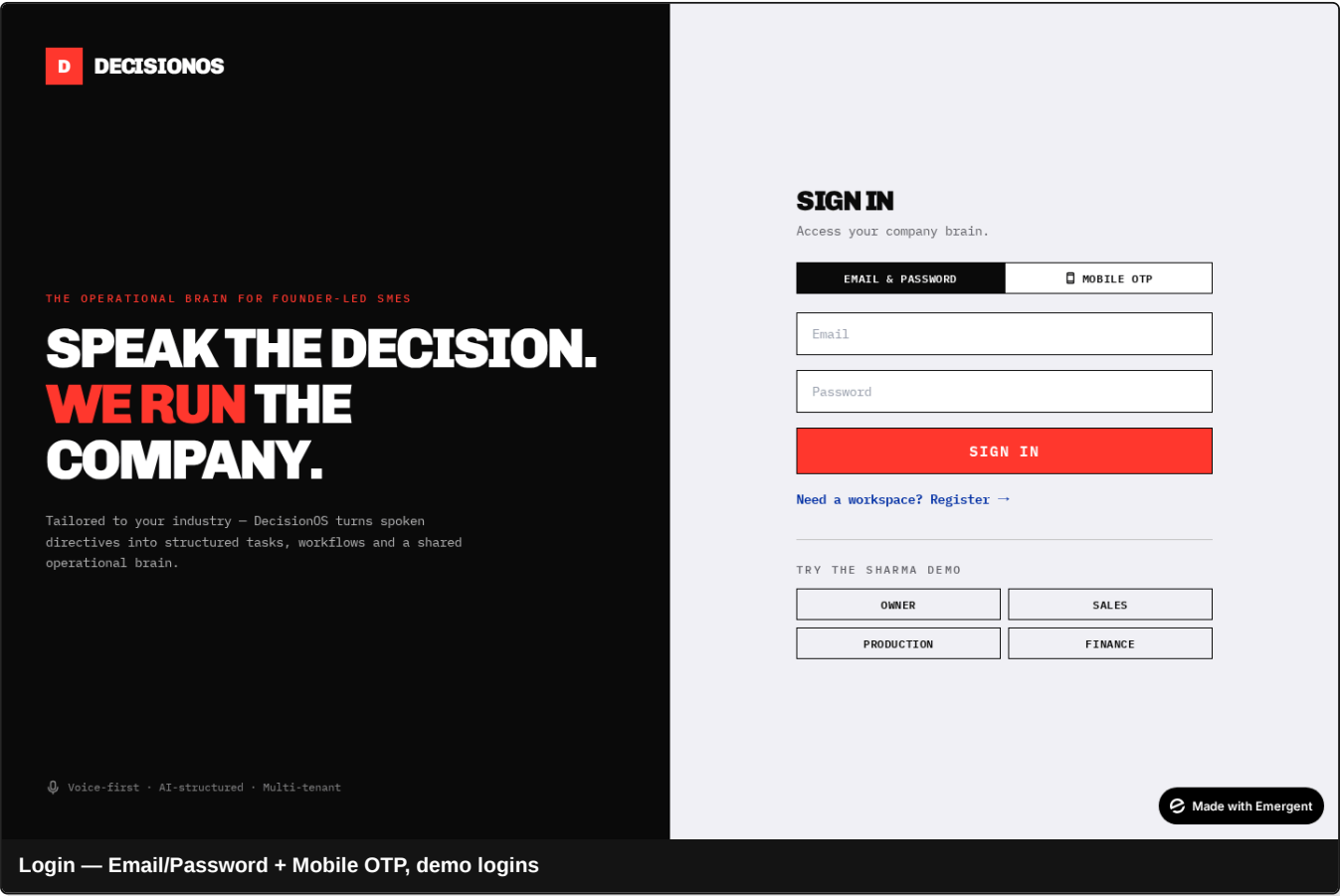
- Owner: owner@sharma.com / demo1234
 - Sales: sales@sharma.com • Production: production@sharma.com • Finance: finance@sharma.com (all demo1234)
 - OTP demo phones (dev): Owner 9820010001, Sales 9820010002, Production 9820010003, Finance 9820010004.
-

12. Roadmap (pending)

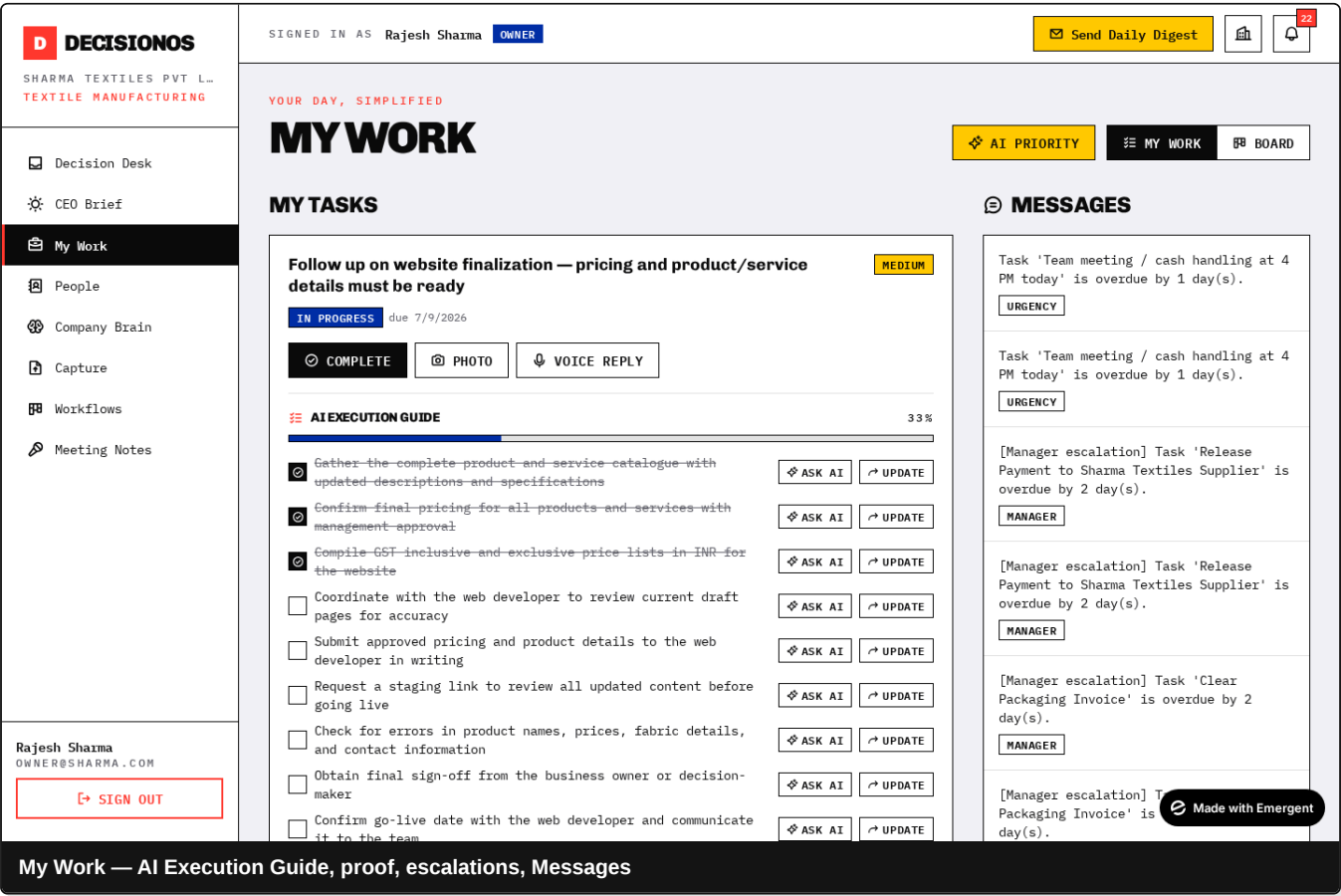
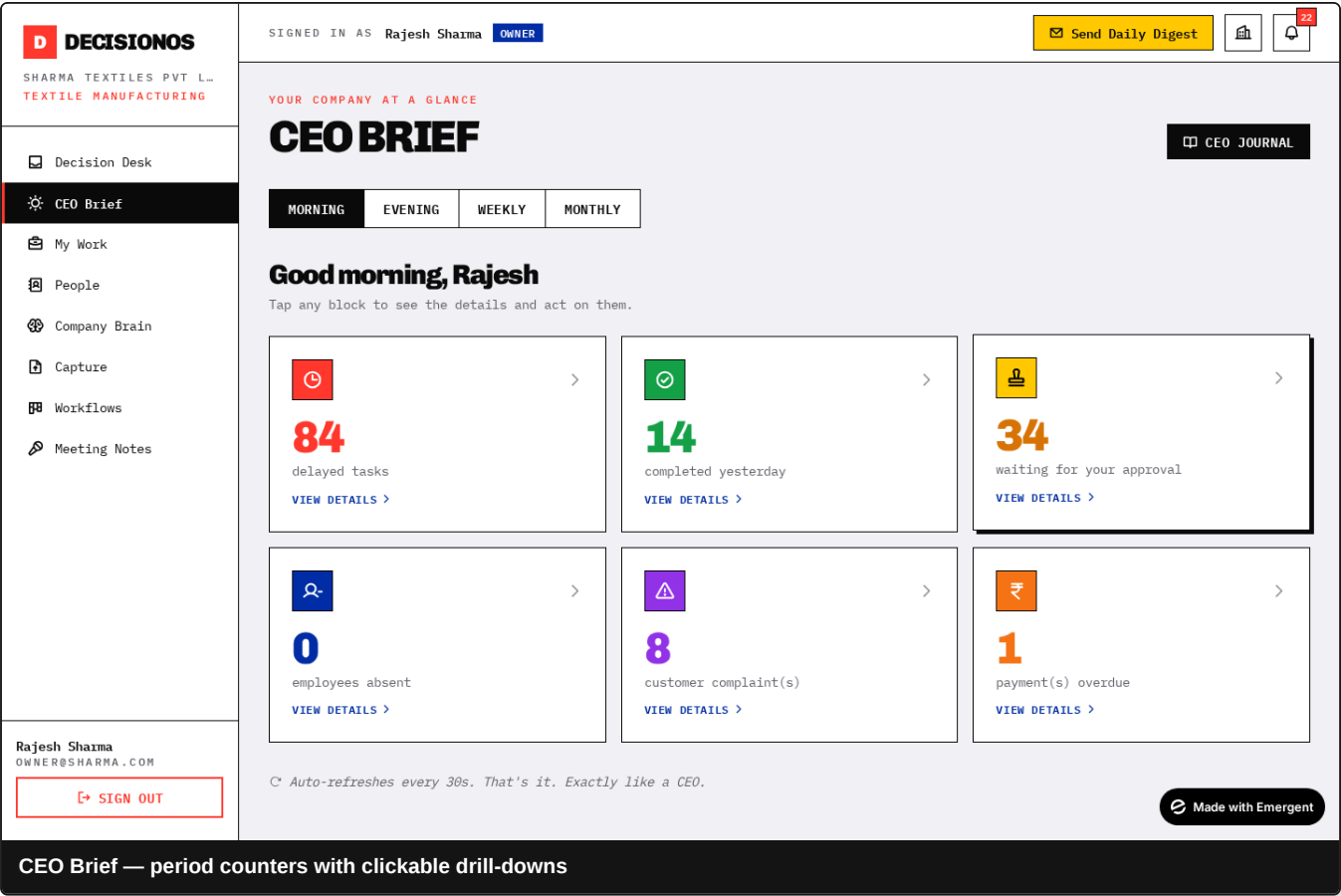
- **P0/P1 — Real SMS:** enable APM Technologies (needs verify/send endpoint) or Twilio → auto-deliver invite/login codes.
 - **P1:** WhatsApp Business Cloud ingestion, Zoho Books sync, Resend email digest (all need keys).
 - **P2:** Tally read-only connector (local agent).
 - **Enhancement idea:** "Pending invites" widget on CEO Brief (who's invited vs. activated, one-tap resend) to close the onboarding loop.
-

Document generated from the live application (Sharma Textiles demo). Desktop and mobile screenshots accompany this document in the chat.

APPENDIX A — VISUAL TOUR (DESKTOP)



Decision Desk — voice/text capture & decision approvals



D

DECISIONOS

SHARMA TEXTILES PVT L.

TEXTILE MANUFACTURING

Decision Desk

CEO Brief

My Work

People

Company Brain

Capture

Workflows

Meeting Notes

Rajesh Shazma

OWNER@SHARMA.COM

SIGN OUT

SIGNED IN AS Rajesh ShazmaOWNER

Send Daily Digest

22

YOUR PEOPLE – TEAM, CUSTOMERS & VENDORS

PEOPLE

EMPLOYEESCUSTOMERSVENDORS

ADD MEMBER

MEMBERS

R

Rajesh Shazma

owner@shazma.com

OWNER

P

Priya Nair

sales@shazma.com

7 ACCESS

SALES

ACCESS

INVITE

MARK ABSENT

A

Amit Verma

production@shazma.com

7 ACCESS

PRODUCTION

ACCESS

INVITE

MARK ABSENT

S

Sunita Rao

finance@shazma.com

8 ACCESS

FINANCE

ACCESS

INVITE

MARK ABSENT

L

Lim

limited1783825982@example.com

2 ACCESS

SALES

ACCESS

MARK ABSENT

Made with Emergent

People — Employees / Customers / Vendors, RBAC + invite links

D

DECISIONOS

SHARMA TEXTILES PVT L.

TEXTILE MANUFACTURING

Decision Desk

CEO Brief

My Work

People

Company Brain

Capture

Workflows

Meeting Notes

Rajesh Shazma

OWNER@SHARMA.COM

SIGN OUT

SIGNED IN AS Rajesh ShazmaOWNER

Send Daily Digest

22

EVERYTHING YOUR BUSINESS KNOWS

COMPANY BRAIN

ASKSEARCH

> DecisionOS Ask AI – grounded in your company data.

> Try one of the queries below to begin.

What purchases need my approval?

Which tasks are overdue?

Summarise open sales orders

What did I decide about festive stock?

> Ask anything about your operations...

ASK

Made with Emergent

Company Brain — Ask (RAG) + Search

Decision Desk

CEO Brief

My Work

People

Company Brain

Capture

Workflows

Meeting Notes

Rajesh Sharma

OWNER@SHARMA.COM

SIGN OUT

SIGNED IN AS Rajesh SharmaOWNER

Send Daily Digest

22

DATA INPUT

INGEST

VIEW RECORDS

PDF

PDF / PHOTO

Invoice, bill, receipt or PO – AI reads it with OCR

CSV / EXCEL

Customer, vendor, sales or payment list – columns auto-detected

WhatsApp forwarding

COMING SOON

Soon your team can forward an invoice or payment screenshot to DecisionOS on WhatsApp and it files itself – powered by this same pipeline.

RECENT UPLOADS

IMG_6190.jpeg

Sales invoice from Ravi kumar to Tanmy sharma for 1028.97 INR, dated 2026-05-22, paid in full.

FILED

2 C · 1 I · 1 P · 0 T

t.csv

1 contact record found with minimal data (name and phone only).

FILED

1 C · 0 I · 0 P · 0 T

t.csv

Customer contact list with 1 record.

FILED

1 C · 0 I · 0 P · 0 T

test_iter12.csv

Customer contact list with 2 entries: TEST_Inbox Ravi and TEST_Inbox Meena.

FILED

2 C · 0 I · 0 P · 0 T

invoice.png

Sales invoice from Sharma Textiles Pvt Ltd to Kumar Garments for Rs 28000.

FILED

0 C · 1 I · 0 P · 1 T

invoice.png

Sales invoice INV-2045 from SHARMA TEXTILES PVT LTD to Kumar Garments for 28000 INR.

FILED

0 C · 1 I · 0 P · 1 T

customers.csv

A customer list containing 3 contacts from Tamil Nadu with GST numbers and contact details.

FILED

0

Made with Emergent

Capture — PDF/Photo OCR + CSV/Excel ingestion

Decision Desk

CEO Brief

My Work

People

Company Brain

Capture

Workflows

Meeting Notes

Rajesh Sharma

OWNER@SHARMA.COM

SIGN OUT

SIGNED IN AS Rajesh SharmaOWNER

Send Daily Digest

22

FLAGSHIP OPERATIONAL FLOWS

WORKFLOWS

+ NEW

ORDER FULFILMENT

PROCUREMENT

ORDER RECEIVED

0

CONFIRMED

37

IN PRODUCTION

16

READY

1

DISPATCH

0

TEST_Order_ADV

TEST_cp

₹500

ADVANCE →

TEST_Order_XYZ

TEST_customer

₹1,000

ADVANCE →

TEST_back

TEST

₹100

ADVANCE →

TEST_Order_ADV

TEST_cp

₹500

ADVANCE →

TEST_back

TEST

₹100

ADVANCE →

TEST_Order_ADV

TEST_cp

₹500

ADVANCE →

TEST_Order_XYZ

TEST_customer

₹1,000

ADVANCE →

TEST_back

TEST

₹100

ADVANCE →

Order #4821 – Delhi Retailer (500 units)

Kapoor Retail Pvt Ltd

₹385,000

ADVANCE →

Made with Emergent

Workflows — Order Fulfilment & Procurement kanban

DECISIONOS

SHARMA TEXTILES PVT L...

TEXTILE MANUFACTURING

Decision Desk

CEO Brief

My Work

People

Company Brain

Capture

Workflows

Meeting Notes

Rajesh Shazma

OWNER@SHARMA.COM

SIGN OUT

SIGNED IN AS Rajesh ShazmaOWNER

Send Daily Digest

22

HOW WELL THE BUSINESS IS RUNNING

OPERATING SCORE

19

/ 100

COMPANY HEALTH

EXECUTION

FINANCE

SALES

RESPONSIVENESS

9

0

77

0

TASKS DONE

71

OPEN TASKS

244

OVERDUE

84

OPEN COMPLAINTS

8

TEAM EXECUTION

1

Rajesh Shazma

OWNER · 43 DONE · 33 OPEN · 1 OVERDUE

COACH

55

2

Priya Nair

SALES · 22 DONE · 101 OPEN · 46 OVERDUE

COACH

0

3

Amit Verma

PRODUCTS...

COACH

0

Made with Emergent

Operating Score — company health + team leaderboard (owner)

APPENDIX B — VISUAL TOUR (MOBILE)

